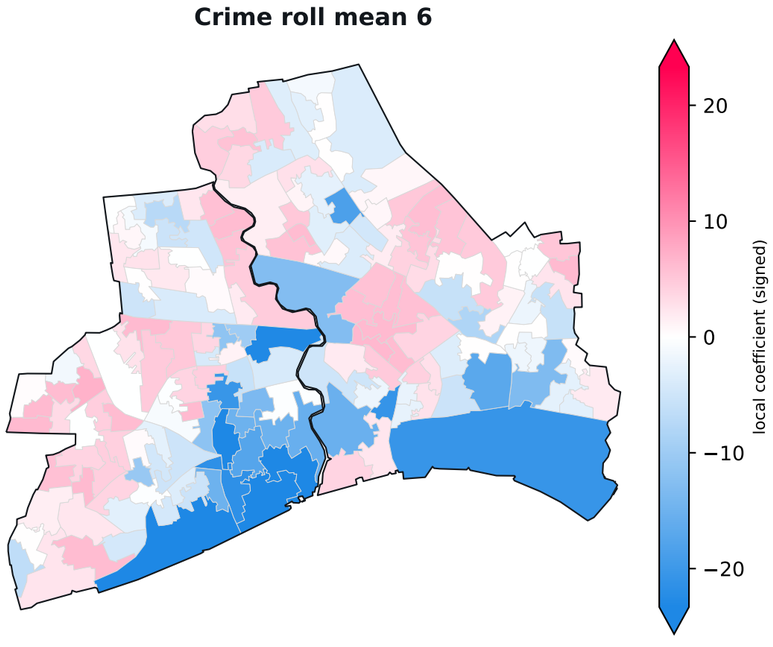
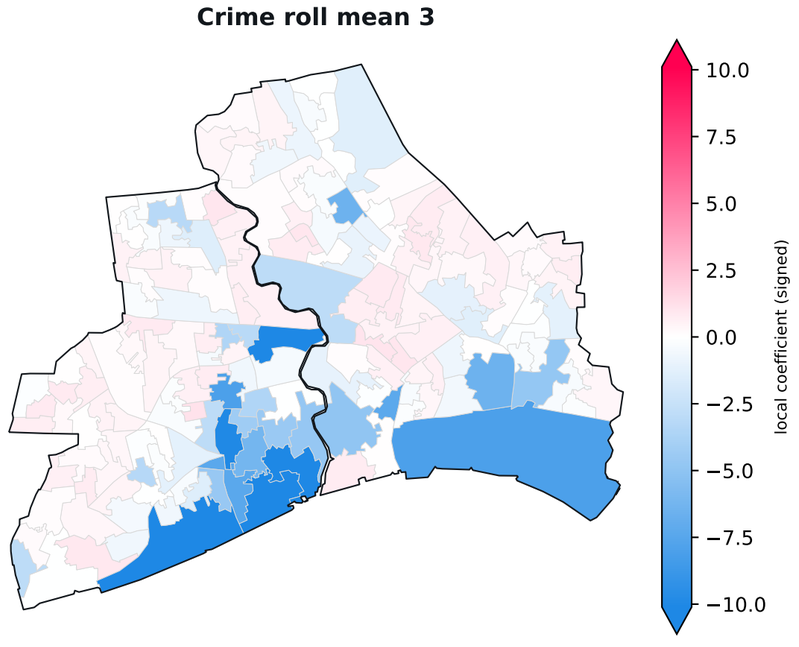
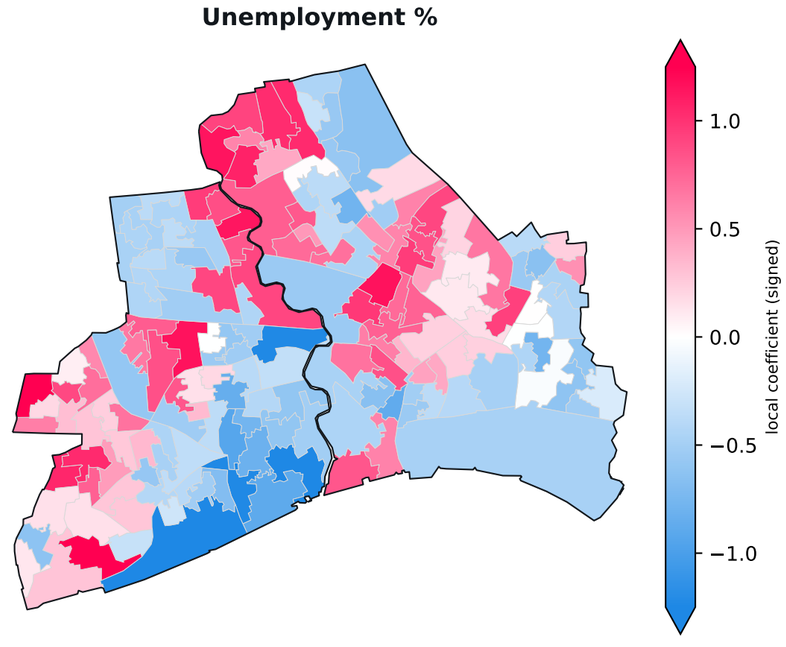






Aligned design: XGBoost on 30 predictors + BNG easting/northing; Feature masking keeps true coordinates; Geo masking moves only an LSOA's coordinates to the training-median point. N=4,000, seed 42. Six leading global Feature drivers; signed local coefficients (original minus perturbed prediction), diverging scale. Not causal effects.